

Learning Chinese History through Digital Game

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Abstract Digital game-based learning is becoming a new trend for teaching and learning. While it has been developing in some countries for years, it is still a relatively new topic in others. With the rapid growth of needs in using information technology in the classrooms and the increasing popularity of digital games, this paper aims at looking into the current situation with the help of a case study.

I. INTRODUCTION

Digital game-based learning has been argued by a lot of different researchers to be one of the most effective ways for the 21st century students to learn [1]. Teenagers, and even adults, nowadays, spend a great period of time on computer and video games. According to [1], average college graduates in the United States spend over 10,000 hours playing video games. Many said that their kids are addicted, and this deeply affects their lives. Yet some also claimed that people are actually learning different aspects of skills and knowledge from digital games [2]. There are a lot of researches that show a strong correlation between digital games and problem solving skills [3], [4], [5], [6], [7], [8], [9]. Most of these studies conclude that problem-solving skills should be improved by playing digital games. Moreover, researches have shown that students are more willing to learn and actually learn better with the help of technology [10].

In the current educational system, history is often regarded as the most boring subject of all [11]. Not only because most of the teachers just read out contents from the boring text book during lessons, but also because these historical backgrounds that build up the world we are experiencing today are in no way related to our modern life [12]. The introduction of digital games and technology would be a good help in this matter [13].

Digital games and technology can transform the traditional classroom into a “constructivist classroom which produced improved skills of question formulating, hypothesis generation and ability to intelligently address new problems” [14]. With chances to get involved and take part in the historical events, digital games are attracting students’ attention and have them concentrated during the classes [10].

With the relationship between digital games and education getting closer and closer, yet not much have been done in the local classrooms. This paper aims at reflecting the local situation by presenting a case study of digital-game based learning in a history class. We will take a look at the possible ways in putting digital games in a history lesson, how the game is designed and produced, how it is

applied into a normal classroom, what kind of impacts does it bring along and what comments do teachers and students give.

II. LITERATURE REVIEW

What students learn at school nowadays lack of real-life context, student are not able to apply what they learnt at school in their everyday life and thus they are not interested in the school subjects [12]. It has been increasingly difficult to capture the interest and attention of young people especially in a traditional classroom setting. The single feed of information and step by step logics just simply can not satisfy the needs of the new generation [15], [16].

In modern schools, History is presented as facts that are unarguable and not in anyhow related to our social life. In [12], Siexans argues that what passes as history – the memorization of a collection of facts, causal explanations and sanctioned narratives – might better be described as the construction of myth or heritage than doing history in any real sense. Treating history as myth or heritage, a “cultural literacy,” tends to lead to many misconceptions about the nature of history, alienate students, and as a result, leave most students unaffected, or even hostile towards school-based history [17], [11], [12], [18]. The majority of students perceives history which is cleaved from social context as the meaningless recitation of names, dates, and facts and fail to understand the practices through which historians construct and judge historical arguments [12], [18]. As a result, unsurprisingly, most students list history as “the most boring” subject studied in school and thus feel no interest in class [11].

In order to adapt to the new life style and learning habits of the students, In [13], Hostetter and Clemens suggest that games should be used in education because that is the way how students are learning now. In [2], Garcia has identified the characteristics of digital games which fit very well with the new generations as follows:

- Speed
- Parallel information processing
- Active engagement
- Quick adaptation and problem solving
- Immediate reward

Studies also show that digital games encourage engagement with complex decisions and offers continuous and instant responsive feedback. Discussion and conversation will be simulated as a result [19]. They also provide a fun, challenging, relaxing, engaging and

motivating experience [20]. Students are more willing to pay attention, participate and more receptive to learning especially when their stress level remains low [10].

Researchers have argued what exactly can be learnt from digital games and the result varies depending on the type of game, the purpose and the site of learning. A lot of parents and teachers have recognized that valuable skills can be develop through playing game, such as, strategic thinking, planning, communication, negotiating skills and group decision making [21]. Researchers have also suggested digital games support the development of logical thinking and problem solving skills [4], [9]. It is a good tool for developing skills required in different stages in the problem solving process: understanding the problem, characterizing the problem, and solving the problem [3]. Collaboration around digital games has been suggested to create a very positive effect on problem-solving [22].

III. CASE STUDY

History teachers from various secondary schools are interested in developing a game for their students. They noticed that students found that History is a boring subject. Plus, the syllabus doesn't cover all the historical incidents that they would like to share with the students. They wanted to bring up details of the three Mongolian conquests to their students in a fun and interesting way. And that's the point when they co-operated with us to develop a digital game that can fulfil these goals. After a series of discussions on the background of the project, we have applied the Incentive Scheme for Developing Instructional Software Funding from the HKSAR Education Bureau for support before starting the production process.

A. Local situation

In order to understand the local situation on the habits and opinions of digital game-based learning, a survey has been carried out among local students, teachers and people working in the field of I.T. in education. The survey was carried out on 21st May, 2009. A total of 40 students, 37 males and 3 females aged between 15 and 20 were being interviewed. 65% of the students have been playing computer games for three or more years in which 30% of them have been engaged in it for even 6 to 10 years.

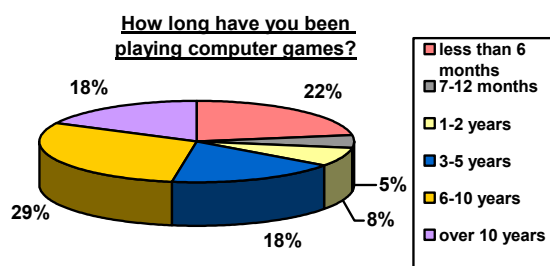


Figure 1. Result of the question "How long have you been playing computer games?"

At the moment, 90% of the interviewees play computer game at least once per week. Half of them spend 1 to 5 hours on these games every week, while 10% of them spend over 20 hours on them. This shows that computer games are really popular among youngsters nowadays and it has been adopted as part of their personal and social way of life.

When asked about their opinion on digital-game based learning, 70% of the students think that they can learn from computer games, yet half of them have never used computer games as a learning tool. For those who have tried using computer games to learn, most of them try it at school or simply by themselves in the form of self-learning.



Figure 2. Result of the question "Do you think you are learning from computer games?"

When asked if they think that playing computer games is a good way to learn, more than half of the students, 60% of them said yes. Furthermore, half of them think that the best situation to do so is at school, followed by learning among friends and then self-learning.

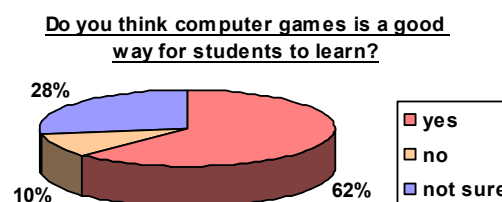


Figure 3. Result of the question "Do you think computer games is a good way for students to learn?"

This survey gives us a basic background of the local situation. It shows that the 21st generation grows up in a society surrounded by technology, especially games. They like playing digital games a lot. They are open to the idea of using computer games as a learning tool, yet not many of them have tried it before. In short, they believe it is a good idea to use digital games to teach and learn at school.

B. Design and production

At the beginning stage of the project, the teachers provided us with content information of the Mongolian conquests that they would like to cover in the game. Game designers and gaming professionals then talked to the teachers to understand the goal of the game and what do

they want the students to achieve after playing. They think about possible game plays for the game and after a series of discussion they had decided to set Mongolian battle field as the gaming background.



Graphic 1. In-game screenshot of Mongolian Conquests

Researches on the graphics of Mongolian scenes and army had been done. Historical information of the three Mongolian conquests was being studied. The game designers worked on the design of the game play while a team of programming professionals built up the game based on the design and a team of artists created appropriate graphics for the game play.



Graphic 2. In-game screenshot of Mongolian Conquests

During the game, players are asked to build up his or her own armies to conquest other countries based on the real Chinese and western history. The player acts as the head of an army. He has to recruit soldiers, feed them, train them, assign them tasks and then when it's time he has to fight with the enemies to conquer the land. Historical information of the three Mongolian Conquests would be brought up from time to time according to the flow of the game play. Students can learn and understand History while they are enjoying the game.



Graphic 3. In-game screenshot of Mongolian Conquests

There is a save feature in the game. Students are allowed to save their game each time so that they can continue playing their game in the next lesson. If the students are interested to know more about the Historical incidents that they come across in the game, after-game materials are also provided. A website has been built for this game in which there are background, instructions and other basic information of the game. Moreover, there is detailed information of the three Mongolian Conquests, together with the people and places that were involved. This gives the chance for students to read about the Historical incidents in detail before or after playing the game.



Graphic 4. Mongolian Conquests supplementary material

C. Applying the game into the classroom

After a one-year production period, the game was finished and being launched onto HKedCity (www.hkedcity.net), a well-known educational resources website in Hong Kong. Teachers and students are able to play the game or use it as a teaching tool in the classroom for free. The game is applicable to most of the computers and technical support is provided to schools whenever needed.

The supplementary webpage was also launched at the same time. Players can go visit the site for basis information of the game. Detailed historical background of the three Mongolian conquests is also provided for the students. Moreover, suggested teaching plans are open to the teachers. They suggest ways that the game can be used in a normal classroom. Reflective questions and activities are also provided for after-game discussions.

History teachers are encouraged to use this game to introduce the three Mongolian Conquests to their students. The game is best to be used in a computer room, with a computer assigned to each of the students. Students are allowed to spend 15-20 minutes on the game in each session. Students are able to save the game and continue playing lesson after lesson. The teacher would do recap and debriefing with the students each time after playing and the students are encouraged to discuss who they have met and what they have experienced in the game. This is to reinforce what the students have learnt about the three Mongolian Conquests each time after playing.

IV. FEEDBACKS AND COMMENTS

Up till now, the game has been available on HKedCity for a year. To find out the usage rate, a set of reports has been generated. Here are some of the key figures: play count for the full version is 6336; play count for the 1st conquest is 1329; play count for the 2nd conquest is 395 and the 3rd conquest has been played 353 times.

In order to get direct feedback and comments of the game, a survey has been distributed to 428 secondary school students and their teachers among 3 secondary schools who have used the game as a teaching tool in the classroom.

A. Students

A total of 428 students have completed the survey, in which 58% of them are males and 42% of them females. 90% of the students are aged between 12 and 14 while the other 10% are aged between 15 and 18.

The first question asked the students whether they like this game as a whole. 83% of the students answered that they like it, with the rest of them saying that they are not sure about it. None of them dislike the game. The next question asked the students if the game gives them a better learning motivation. 74% of them said yes as they learn while they play and have fun.



Fig. 4 Result of the question "Does this game give you a better learning motivation?"

Moreover, 70% of them think that they did learn History from playing this game while the rest of them, 17% said that they didn't and 13% of them are not sure about it. It reflects that after-game assessment and reinforcement are essential to the students' learning. If the teacher can spend some time to talk about what they have learnt during the game, the students would be able to understand it better and thus can strengthen their memory.



Fig. 5 Result of the question "Did you learn history from the game?"

In the next question, the students are asked whether they like this game-based learning approach. 81% of them claimed that they like it while 7% of them are not sure about it. It reflects that most of the students accept and prefer using games to learn. In the last question, students are asked whether they think they learn better under this game-based learning approach. Though it might be the first time for most of them to experience this mean of learning, more than half of them think that they do learn better this way. Some students' comments on Mongolian Conquests include "it attracts me to play", "it is nice to learn in a happy way", "we like games, so education mixing games is good.", most of the students are playing computer games at home, it is good that if we play this educational game", "it is easier for me to learn through games".

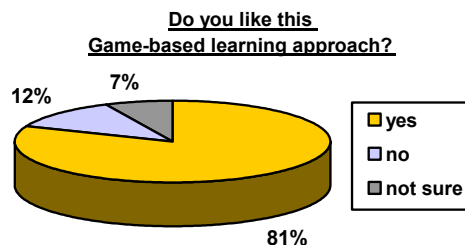


Fig. 6 Result of the question "Do you like this Game-based learning approach?"

In short, students like this game. They think that the game gives them a better learning motivation and that they are learning the subject matters from the game. Nearly all of the students like this game-based learning approach and they do believe that they can learn better with the help of digital games.

B. Teachers

As for the teachers, it was not the first time for them to use computer games in the classrooms. This time, they let their students spend around 20 minutes on the game in each lesson and they found out that there are positive changes in the students' learning attitude. The students felt much more interested in class and were more concentrated as well. The teachers think that the game is a good one and they like using it as a teaching tool. In general, the teachers taking this survey support game-based learning and will continue adopting this approach in their classes.

V. CONCLUSION

We understand that students are playing digital games; students can learn from digital games and that teachers find digital games useful in teaching. Digital-game based learning is an ongoing and growing discussion topic among educators and gaming professionals. It is not new but there are still a lot to explore.

Lessons have been given in the same way for the last 30 years. Students are told to keep quiet and listen to the teachers in the classrooms. What are being taught and how they are being taught can't arise students' interests. More interaction, more involvement and more challenges are required not only to enrich student's textbook knowledge but also to equip them with skills that they need to acquire to survive in the 21st century. At this point, the implement of technology in the classrooms is essential. And teaching through digital games seems to be one of the most popular ways among teachers and students.

Digital games encourage players to observe what is going on, to analyze the situation they are under and to decide how to react in getting the best results. There are numerous possibilities during the game play and the ending is for the players to decide. Players are in total control of their experience. This is also the most natural way to learn. Unfortunately, that's not how students learn in the classroom. Even if we don't like digital games, maybe there are some ideas that we can borrow from them. We can also empower students to be designers, let them create their own games for learning. Under a team-based environment, personal and social skills can also be developed.

Using this case study as an example, we hope to provide a general picture of the process of creating a digital game and the possible way in bringing it into the classroom. It includes the early stage of teacher's realization of using digital game as a teaching tool, the ideas of game creation, the process of production and finally the application of the game onto the students. More detailed and formal researches are needed in this area yet we hope this example could show how digital games could benefit the teaching and learning process. We seek for advices and improvements while looking for a possible future of digital game-based learning in the classrooms.

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